

Intergenerational Housing Misallocation and Fertility

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Housing and Fertility

Children increase how big of a house you need. This has implications for:

- Renting: large rental units are scarce.
- Owning: a family-sized home requires liquid wealth and a down payment.
- Wealth: older households retain much of the large owner stock after children leave.

Physical space is key. How much does it affect fertility?

What kind of housing policy can improve access to family-sized homes?

→ focus on space constraints and tenure choice

This Paper

Need a structural model where fertility, tenure, housing size, and saving are joint.

First: a simplified two-generation model

- Isolate how rental limits, down payments, and old-owner retention affect the shadow value of space.
- Characterize the fertility response and the gain from reallocating family-sized housing.

Then: a full quantitative lifecycle model

- Add income risk, saving, tenure adjustment, transaction costs, bequests, entry, and housing supply.
- Discipline the model with fertility, housing, tenure, and wealth moments.

Finally: use the full model to evaluate policies that change access when households make fertility decisions.

I. Simplified Analytical Model

Isolating access, fertility, and intergenerational allocation

Environment

Young households

- Draw income and liquid wealth, $i = (y_i, b_i)$.
- Choose entry, tenure, housing, completed fertility, and saving.
- Become old owners or old renters next period.

One stock $\bar{H}$ clears at rental user cost $r = \rho P$. Renting is capped at $\bar{h}^R$ and owning at $\bar{h}^O$, with $\bar{h}^R < \bar{h}^O$.

Old households

- Owners choose how much housing to retain and the estate to leave.
- Renters demand housing without an ownership lock-in wedge.
- All households exit after old age.

Preferences

A young household receives utility from adult consumption, residual space, and completed fertility:

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

- Each child requires χ units of consumption and κ units of housing.
- The cost of a child therefore depends on the household's ability to access space.
- Old households value consumption, retained housing, and a warm-glow estate.

Markets and Housing Technology

One fixed stock $\bar{H}$ of divisible floorspace can be rented or owned.

$$h \leq \bar{h}^R \quad \text{if renting,} \quad h \leq \bar{h}^O \quad \text{if owning,} \quad \bar{h}^R < \bar{h}^O.$$

- Family-sized space is available only through ownership.
- A single market clears, and no arbitrage links rent and the asset price:

$$r = \rho P, \quad \rho = q^{-1} - 1 + \delta + \tau^P.$$

- A buyer finances at most share ϕ , so the down payment is $(1 - \phi)Ph$.
- Selling realizes a nominal capital gain; death receives stepped-up basis.

The Young Household's Problem

Conditional on entry, household i chooses tenure $m \in \{R, O\}$ and solves

$$\begin{aligned} W_i^m = \max_{c_i, h_i, n_i, b'_i} & u_i^Y(c_i, h_i, n_i) + \beta V^m(b_{j'}, h_i) \\ \text{s.t.} \quad & c_i + rh_i + b'_i = y_i + b_i, \\ & h_i \leq \bar{h}^m, \quad b'_i \geq 0, \\ & \mathbf{1}\{m = O\}(1 - \phi)Ph_i \leq b_i. \end{aligned}$$

- Tenure is chosen by comparing W_i^R and W_i^O .
- Entry compares the best inside value with an outside option under extreme-value shocks.
- Current liquid wealth determines whether the ownership branch is feasible.

Old Owners and Lock-In

An old incumbent enters with liquid wealth b_j , a house H_j^0 , and an accrued gain:

$$\begin{aligned} V^O(b_j, H_j^0) = \max_{c_j^2, h_j^2, e_j} & \log c_j^2 + \gamma \log h_j^2 + \omega_B \log e_j \\ \text{s.t.} \quad & c_j^2 + e_j + r h_j^2 = b_j + r H_j^0 - \bar{\ell}(H_j^0 - h_j^2). \end{aligned}$$

- Selling a unit nets $r - \bar{\ell}$ in flow-equivalent terms.
- Keeping it until death avoids the capital-gains tax through stepped-up basis.
- At an interior choice, the old incumbent values marginal retained space at $r - \bar{\ell} < r$.
- Old renters carry no tax wedge and value marginal space at r .

Stationary Competitive Equilibrium

A stationary equilibrium is a rental rate r , asset price $P = r/\rho$, household policies, entry probabilities, and stationary measures such that:

1. young households choose entry, tenure, fertility, housing, and saving optimally;
2. old owners and renters choose housing and estates optimally;
3. young owners become old incumbents and young renters become old renters;
4. births plus the renewal inflow reproduce the next potential-young cohort;
5. the single floorspace market clears:

$$\bar{M} \int \pi_i^E h_i dG_Y(i) + \int h_j^2 dF_O(j) + \int h_j^2 dF_R(j) = \bar{H}.$$

Market clearing determines the price, but not whether family-sized space reaches the household that values it most.

Housing, Tenure, and the Effective Cap

Using $P = r/\rho$, the maximum housing available in each tenure is

$$H_i^R = \bar{h}^R, \quad H_i^O = \min \left\{ \bar{h}^O, \frac{b_i \rho}{(1 - \phi)r} \right\}.$$

- Renters are constrained by the largest available rental unit.
- Owners may be constrained by the physical menu or by the down payment.
- A household can afford the flow cost of space yet remain unable to purchase it.

The relevant friction is therefore not tenure by itself, but the household-specific cap on family-sized space.

The Access Gap

Divide the housing first-order condition by the marginal utility of consumption. For a constrained renter,

$$\frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = r + \underbrace{\frac{\theta_i}{\lambda_i}}_{\zeta_i^R \geq 0}.$$

For an owner, the gap separates financing from the physical size limit:

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i}(1 - \phi)P}_{\text{down-payment margin}} + \underbrace{\frac{\theta_i^O}{\lambda_i}}_{\text{owner size limit}}.$$

- ζ_i^m is the goods value of relaxing the household's constraint by one unit of space.
- It can be read from the household's consumption and housing bundle, net of child needs.

The Full Price of a Child

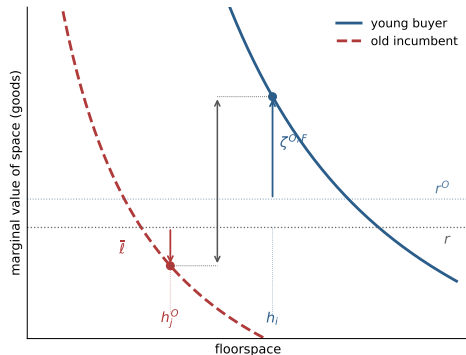
The fertility first-order condition prices a marginal child in units of goods:

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a child}} = \underbrace{\chi + \kappa(r^m + \zeta_i^m)}_{\text{full price of a child}}.$$

- χ is the child's direct consumption requirement.
- κr^m is the market cost of the space required by the child.
- $\kappa \zeta_i^m$ is the additional cost created by limited access to that space.

Interpretation: a binding housing constraint acts like a household-specific tax on children even when the market user cost does not change.

Intergenerational Misallocation

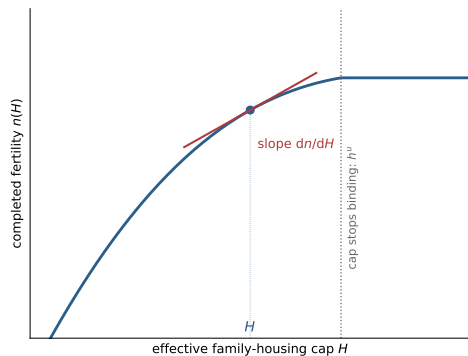


Moving one unit from a locked-in old incumbent to a financing-constrained young buyer produces surplus

$$\zeta_i^{O,F} + q\bar{\ell} + \bar{\ell} > 0.$$

- The buyer values space above effective user cost.
- The incumbent values retained space below the market rent because selling realizes a taxable gain.
- The common market price cancels.

Fertility and Policy



Along a binding cap, more space has two effects:

- it makes the child's housing requirement easier to meet;
- it tightens the budget through the cost of additional space.

Fertility rises when the housing channel dominates:

$$\chi \leq (1 + \Lambda) \frac{\kappa r^m}{\alpha}.$$

Exposure and pass-through determine the aggregate effect of a housing policy.

Policies Relax Different Housing Constraints

The effective owner cap is

$$H_i^O = \min \left\{ \bar{h}^O, \frac{b_i \rho}{(1 - \phi)r} \right\}, \quad H_i^R = \bar{h}^R.$$

Policy	Directly relaxed margin
Larger rental units	Rental cap $\bar{h}^R$
Down-payment grant	Liquid wealth b_i at purchase
Looser mortgage credit	Financed share ϕ
Property-tax change	Asset price $P = r/\rho$ and required equity

A policy has a direct fertility effect only for households at the margin it relaxes. The aggregate response also depends on exposure, pass-through, entry, and tenure switching.

What the Simplified Model Delivers

- Rental scarcity and down-payment constraints combine into a household-specific cap on family space.
- The access gap ζ_i^m measures the marginal value of relaxing that cap.
- The same gap enters the full price of a child through $\kappa\zeta_i^m$.
- Reallocating space from a locked-in old owner to a constrained young buyer can generate positive surplus.
- Fertility rises when relaxing the housing requirement dominates the associated resource cost.

The simple model identifies the mechanism. The quantitative model asks how many households are exposed, how large the gaps are, and how prices and population adjust.

II. Full Quantitative Lifecycle Model

Measuring the mechanism and evaluating policy in equilibrium

Quantitative Lifecycle Model

18	18–42	66	85
entry	fertile window	retirement	maximum age

- Households face persistent earnings risk and save in liquid assets.
- A childless household makes a one-time discrete completed-family-size choice; children then age and leave home.
- Each period households choose renting or one of five owner sizes and adjust saving.
- Entry and the stationary population scale respond in counterfactuals.

The model preserves the analytical mechanism while adding the lifecycle margins needed for measurement and policy.

Lifecycle and Income

Before retirement, household income combines an age profile and persistent idiosyncratic risk:

$$y(a, z) = \begin{cases} (1 - \tau_{\text{pay}}) w e_a z, & a \leq J_R, \\ p^{\text{ret}}, & a > J_R, \end{cases} \quad z' \sim \Pi^z(\cdot | z).$$

- One period is four years; households enter at 18, retire at 66, and live at most to 85.
- e_a is the deterministic lifecycle earnings profile.
- Annual log income has persistence 0.960 and innovation standard deviation 0.0645.
- The process is sampled every four years and approximated with a five-state Rouwenhorst chain.
- Survival is certain before retirement and age-specific thereafter.

Preferences and Family Needs

Dependent children shift minimum consumption and housing requirements:

$$\bar{c}(n, s) = \bar{c}_0 + \bar{c}_n n_s, \quad \bar{h}(n, s) = \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s.$$

Flow utility is CRRA over residual consumption and usable housing services:

$$u = \frac{[(c - \bar{c})^{\alpha_c} \mathbf{h}^{1-\alpha_c}]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} n_s,$$

where usable housing services are

$$\mathbf{h}(h^R, h; n, s) = \begin{cases} h^R - \bar{h}(n, s), & h = 0, \\ \chi_O[h - \bar{h}(n, s)], & h > 0. \end{cases}$$

The first child can create a discrete housing jump; additional dependent children increase housing needs linearly.

Bequests

At death, bequeathable wealth is liquid wealth plus the gross value of the house:

$$W^B = b + Ph.$$

Warm-glow utility is

$$\mathcal{B}(W^B, n) = \theta_0(1 + \theta_n n) \frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1 - \sigma}.$$

- θ_0 governs the strength of the bequest motive.
- θ_1 shifts its curvature with wealth.
- θ_n allows completed fertility to change the marginal value of an estate; it is set to zero in the benchmark.
- There is no forced sale or transaction cost at death.

Housing, Tenure, and Budget Constraints

Renters

- Choose rooms continuously up to $\bar{h}^R = 6$.
- Cannot borrow: $b' \geq 0$.
- Pay rental user cost r .

Owners

- Choose $h \in \{2, 4, 6, 8, 10\}$ rooms.
- Can borrow up to $\phi = 0.80$ of house value.
- Pay depreciation, property tax, and a 6% sale cost when adjusting.

Housing supply is upward sloping,

$$H^S(r) = \bar{H} \left(\frac{r}{\bar{r}} \right)^\eta, \quad \eta = 1.75,$$

so higher demand is absorbed by both quantities and prices.

Fertility and Child Aging

A childless household chooses completed family size during the fertile window:

$$n' \in \mathcal{N} = \{0, 1, 2\}.$$

- $n' = 0$ means no birth in the current period; a positive choice fixes completed fertility forever.
- If the household reaches age 42 without a positive choice, it remains childless.
- A positive choice moves children into the first dependent stage.
- Children age stochastically and remain at home for 18 years on average.
- All children belong to one cohort and leave together; the model does not yet allow sequential births.
- Completed fertility continues to matter for bequests after children leave home.

Within-Period Choice Sequence

A household begins with state

$x = (b, z, h, a, n, s)$: wealth, income, housing, age, completed fertility, child stage.

1. If childless and fertile, observe family-size taste shocks and choose n' .
2. Observe housing taste shocks and choose the next owned position

$$h' \in \{0, 2, 4, 6, 8, 10\};$$

$h' = 0$ means renting.

3. Choose consumption and end-of-period liquid assets; renters also choose current rental space.
4. Income and child-stage risk realize before the next period.

A birth can therefore change tenure and occupied space in the same period, rather than only affecting future housing demand.

Renter and Owner Problems

Conditional on renting,

$$\begin{aligned} V^R(\tilde{b}, z, a, n, s) = \max_{c, h^R, b'} & \quad u(c, h^R, 0; n, s) + \beta \bar{V}(b', z, 0, a+1, n, s) \\ \text{s.t.} & \quad c + rh^R + b' = \tilde{b} + y(a, z), \\ & \quad 0 \leq h^R \leq \bar{h}^R, \quad b' \geq 0. \end{aligned}$$

Conditional on choosing owner rung $h' > 0$,

$$\begin{aligned} V^O(\tilde{b}, z, h', a, n, s) = \max_{c, b'} & \quad u(c, 0, h'; n, s) + \beta \bar{V}(b', z, h', a+1, n, s) \\ \text{s.t.} & \quad c + (\delta + \tau^P)Ph' + b' = \tilde{b} + y(a, z), \\ & \quad b' \geq -\phi Ph'. \end{aligned}$$

Renting offers continuous space but no borrowing; ownership offers larger discrete units and collateralized borrowing.

Housing Adjustment and Tenure Choice

Liquid wealth after changing the owned housing position is

$$\tilde{b}(b, h, h') = \frac{b}{q} + \mathbf{1}\{h > 0, h' \neq h\}(1 - \psi^s)Ph - \mathbf{1}\{h' > 0, h' \neq h\}Ph'.$$

- A seller receives the transaction-net value $(1 - \psi^s)Ph$.
- A new buyer pays the full purchase price Ph' and must clear the origination down payment.
- An incumbent who keeps the same house makes no transaction.
- Rent-or-own and owner-size choices have Type-I extreme-value shocks with scale κ_T :

$$\pi^H(h' | x) = \frac{\exp\{\tilde{V}^H(h' | x)/\kappa_T\}}{\sum_{\hat{h} \in \mathcal{H}} \exp\{\tilde{V}^H(\hat{h} | x)/\kappa_T\}}.$$

The Entry Problem

At age 18, a potential household chooses the modeled housing market or an outside economy before drawing initial wealth and income.

$$W^E(r) = \int V(b, z, 0, 1, 0, 0; r) dG_E(b, z).$$

With extreme-value entry tastes,

$$q^E(r) = \frac{\exp\{W^E(r)/\kappa_E\}}{\exp\{W^E(r)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}.$$

- Entrants begin as childless renters.
- Initial (b, z) is drawn from the entry distribution G_E .
- Lower housing costs raise the value and probability of entry.
- Locally born and outside-origin entrants draw from the same G_E .

Entry Equilibrium and Stationary Scale

There are two sources of potential entrants:

- a fixed outside-origin mass M ;
- locally born children who mature, $SB_0(g, r)$.

Stationarity requires entrants to replace the age-one mass needed by the stationary composition:

$$q^E(r) [M + SB_0(g, r)] = SE_0(g, r).$$

Hence

$$S = \frac{q^E(r)M}{E_0(g, r) - q^E(r)B_0(g, r)}.$$

Housing costs affect population through entry $q^E(r)$; fertility affects population through future locally born entrants $B_0(g, r)$.

Stationary Equilibrium and Computation

A stationary equilibrium combines:

1. optimal consumption, saving, fertility, and housing policies;
2. a stationary household distribution $g = \mathcal{T}(g; r, S)$;
3. entry probability $q^E(r)$ and population scale S ;
4. housing-market clearing in levels:

$$S\hat{H}^D(g, r) = H^S(r);$$

5. government accounting for taxes, pensions, grants, and rebates.

I solve the household problem by backward induction, obtain the stationary distribution by forward iteration, and adjust the user cost until housing demand equals supply.

What the Full Model Adds

- A lifecycle of earnings, saving, ownership, children at home, retirement, and death.
- A discrete owner ladder and continuous rental choice, with transaction costs and collateralized borrowing.
- Fertility and tenure decisions made jointly in the same period.
- Bequests and late-life housing retention.
- Endogenous entry, population scale, and upward-sloping housing supply.

The analytical model isolates the access gap. The full model measures it in the stationary distribution and allows policies to change prices, quantities, entry, and fertility together.

III. Quantification

Calibration, model fit, mechanism, and policy

Externally Calibrated Parameters

Block	Parameter	Value or restriction
Lifecycle	Period / entry / retirement / maximum age	4 years / 18 / 66 / 85
Preferences	Risk aversion σ	2
Finance	Real return / financed share ϕ	2.0% annual / 0.80
Housing	Depreciation / property tax / sale cost	1.1% / 1.0% / 6% annual
Income	Persistence / innovation s.d.	0.960 / 0.0645 annual
Products	Owner sizes / rental cap	{2, 4, 6, 8, 10} / 6 rooms
Supply	Housing-supply elasticity η	1.75
Entry	Initial wealth distribution	PSID young childless renters
Children	Expected dependent duration	18 years
Bequests	Child shifter θ_n	0

The income innovation variance is provisional; quantitative results are conditional on this restriction.

Internally Calibrated Parameters

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.9912
α_c	Consumption weight	0.5912
$\bar{c}_0 / \bar{c}_n$	Baseline / per-child consumption need	1.2596 / 0.3929
$\bar{h}_0$	Baseline housing need	0.3915
$\bar{h}_{\text{jump}} / \bar{h}_n$	First-child jump / per-child housing need	1.6021 / 0.1644
ψ_{child}	Utility from children	0.1979
κ_F	Family-size choice dispersion	2.0434
χ_O	Owner service premium	1.1133
$\bar{H}$	Housing-supply scale	8.6456
θ_0 / θ_1	Bequest strength / curvature shift	0.3118 / 0.3973
κ_T	Tenure-choice dispersion	0.0100

Fourteen parameters are estimated jointly; the blocks are an interpretation of identifying variation, not separate estimation steps.

Calibration Targets

Family size

- Completed fertility
- Childlessness
- Family ownership gap
- First-child rooms increment
- Rooms gap: 3+ vs. 1–2 children

Tenure and housing

- Ownership ages 30–55
- Renter rooms
- Large-home share among owners
- Owner–renter rooms gap
- Ownership ages 25–34 and 65–75
- Mean occupied rooms

Wealth and bequests

- Young renter liquid wealth
- Late-life estate median
- Older nonhousing-wealth share

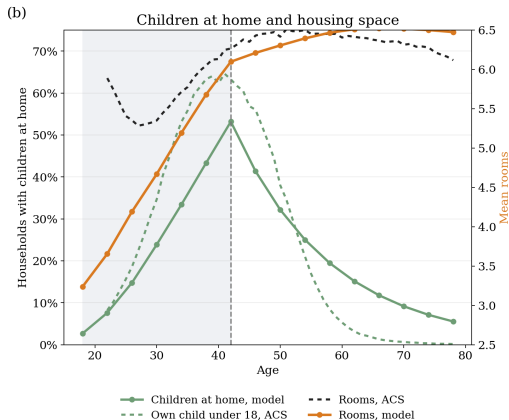
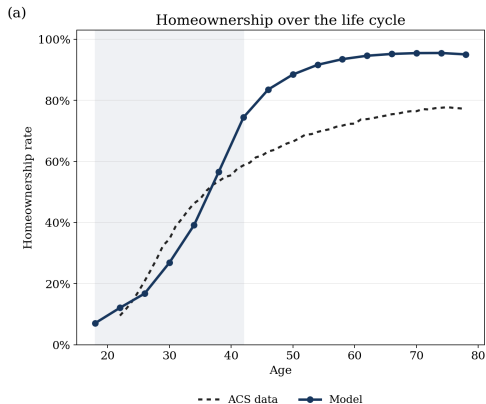
CPS disciplines completed family size; ACS disciplines tenure and rooms; PSID disciplines entry wealth, saving, and estates.

Model Fit

Moment	Data	Model
Completed fertility / childlessness	1.918 / 0.188	2.034 / 0.189
First-child rooms increment	0.664	0.681
Family ownership gap	0.168	0.219
Ownership, ages 30–55	0.575	0.658
Ownership, ages 25–34	0.341	0.274
Ownership, ages 65–75	0.764	0.954
Renter rooms / owner–renter rooms gap	3.805 / 2.419	3.963 / 2.323
Childless-owner share with ≥ 6 rooms	0.596	0.760
Young liquid wealth / income	0.179	0.328
Estate median / income	6.501	6.431
Older nonhousing-wealth share	0.608	0.610

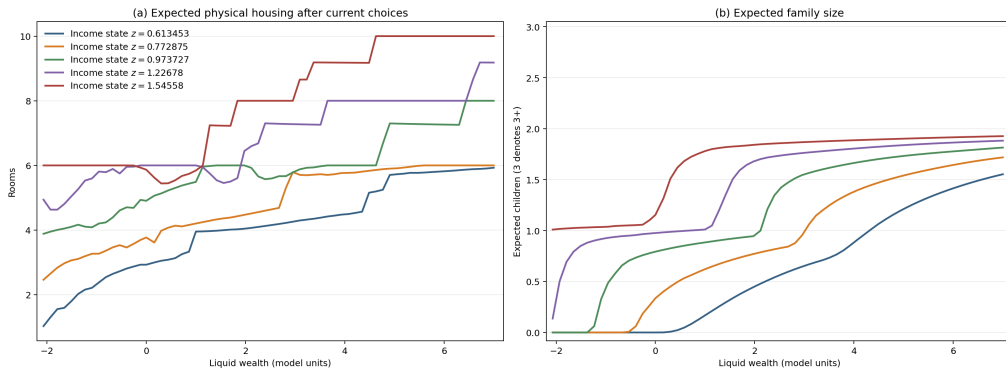
The largest tension is excessive persistence in homeownership: young households buy too late, while older households rarely exit ownership.

Lifecycle Housing Patterns



Homeownership is understated through the mid-thirties and overstated from the early forties onward. The child-at-home profile has the right hump but is delayed and too persistent.

Housing and Family Size at Age 42



At age 42, higher-income households choose children and larger homes at lower liquid-wealth levels. The discrete owner ladder generates sharp changes in physical rooms.

Quantitative Evidence on the Mechanism

Among fertile renters whose positive-family branch reaches the rental cap:

$$\underbrace{r}_{0.130} + \underbrace{\zeta_i^R}_{0.178} = \underbrace{0.307}_{\text{marginal value of another room}} = 2.37 r.$$

Holder of owner homes with ≥ 6 rooms	ACS	Model
Households age 46+	71.8%	80.7%
Households with no children at home	54.9%	74.8%

The households facing the highest marginal value of family space are distinct from those holding most of the family-sized owner stock.

Housing Policy

Policy	Δ total births	Δ housing price
Purchase grant for family-sized homes	+2.1%	+0.90%
Higher property tax plus purchase grant	+2.7%	-18.44%

- The grant covers roughly one-third of the required down payment on an eligible home.
- The package also raises the annual property-tax rate on owner housing from 1 to 2 percent.
- Entry, population, and the housing price adjust in both exercises.

Conclusion

- A model of completed fertility with tenure choice, saving, and a lifecycle of housing need.
- Children raise housing need; rental limits and down payments make space especially costly for young households.
- Older households hold most large owner homes. With retention wedges, a cleared market need not allocate them efficiently.
- Aim is policy: purchase assistance raises births, and the tax-plus-grant package changes both access and prices.

More work is needed: sequential fertility, income risk, old-age downsizing, and across-market relocation.